

Tejas Rohit Kharkar

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Professional Summary

AI and Embedded Systems Engineer with hands-on experience in Edge AI, TinyML, Computer Vision, IoT, ESP32 and STM32 development. Skilled in deploying machine learning models on resource-constrained devices, building real-time sensor-based systems, and developing industrial IoT solutions. Strong foundation in Python, C/C++, OpenCV, TensorFlow Lite and embedded firmware development. Seeking AI/IoT Engineer opportunities focused on intelligent edge systems and real-world AI deployment.

Technical Skills

Programming: Python, C, C++, SQL

AI/ML: Machine Learning, Deep Learning, Computer Vision, Generative AI, TinyML, TensorFlow Lite, OpenCV, Model Optimization

Embedded Systems: ESP32, STM32, Arduino, Sensor Integration, Firmware Development

IoT & Communication: MQTT, BLE, Wi-Fi, I2C, SPI, UART

Tools: Git, Edge Impulse, Linux, VS Code

Experience

AI Engineer — Hashtee Labs

May 2026 – Present

- Developed Computer Vision and Edge AI solutions for industrial automation and monitoring applications.
- Built OpenCV-based image processing pipelines for real-time detection and tracking.
- Integrated AI models with embedded hardware for low-latency inference.
- Implemented MQTT-based communication systems for IoT device management.
- Worked on deployment optimization and model performance tuning.

Technical Lead – Embedded & Edge AI — Shunya

Aug 2023 – Present

- Built 5+ embedded AI and IoT prototypes using ESP32 and STM32.
- Designed complete TinyML pipelines including data collection, labeling, training, quantization and deployment.
- Developed BLE Smart Lock using RSSI-based proximity authentication.
- Built portable health monitoring device measuring SpO2, BPM and temperature.
- Implemented MPU6050-based motion analysis system for tremor detection.
- Optimized firmware for latency, memory footprint and power consumption.

Projects

Edge AI Object Detection on ESP32

- Trained and deployed TinyML model using Edge Impulse.
- Integrated TensorFlow Lite Micro for on-device inference.
- Achieved real-time predictions with OLED visualization.

AI-Powered PPE Detection System

- Developed real-time safety monitoring system using YOLO and OpenCV.
- Detected helmets and safety gear from CCTV streams.
- Optimized inference for deployment on edge hardware.

Smart Industrial Machine Health Monitoring

- Built ESP32-based vibration and temperature monitoring system.

- Implemented anomaly detection for predictive maintenance.
- Published telemetry through MQTT dashboard.

BLE Smart Lock System

- Developed proximity-based authentication system using RSSI.
- Implemented secure embedded decision-making logic.
- Added fail-safe mechanisms and low-power operation.

Assistive Robotics for Parkinson's Patients

- Designed stabilizing spoon system using MPU6050.
- Visualized tremor patterns with embedded processing.
- Provided OLED-based feedback interface.

Multiscope Health Monitor

- Built portable health monitoring platform.
- Integrated SpO2, BPM and temperature sensors.
- Enabled real-time data visualization on OLED display.

Education

B.Tech in Artificial Intelligence

University of Engineering and Management, Jaipur
2023 – 2027

Certifications

- Introduction to Generative AI
- Introduction to Large Language Models
- Tata GenAI Powered Data Analytics
- Responsible AI
- Edge Impulse TinyML Fundamentals

Achievements

- Built and demonstrated multiple working AI and IoT hardware prototypes.
- Participated in hackathons focused on healthcare and smart automation solutions.
- Developed end-to-end Edge AI systems from dataset creation to deployment.